

Jayesh Katade

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EDUCATION

Northeastern University, Boston, MA

Jan 2023 – May 2025

Master of Science in Data Science (GPA: 3.5/4.0)

Maharashtra Institute of Technology, Pune, India

Jul 2018 – Jul 2022

Bachelor of Technology in Computer Science (GPA: 3.6/4.0)

WORK EXPERIENCE

Data Analyst & Teaching Assistant | Northeastern University, Boston

Aug 2023 – May 2025

- Analyzed gradebook data with Pandas and visualized performance trends using Matplotlib, enabling interventions that raised average scores by 22%.
- Tested machine learning approaches in Scikit-learn to predict academic performance trends and flag discrepancies between proctored and unproctored assessments.
- Mentored 3,600 students over 4 semesters in Python by creating coding assignments, hosting office hours, and leading interactive sessions, improving comprehension and engagement by 35%.

Research Data Analyst | Whiz Key Pvt. Ltd., Pune, India

Jul 2021 – Dec 2022

- Evaluated and summarized over 30 research studies and industrial IoT time-series datasets, extracting key methodologies for trend detection and predictive maintenance.
- Tested preprocessing techniques using Python, Pandas, and NumPy to clean and structure simulated sensor data for analysis.
- Created exploratory dashboards in Matplotlib, Seaborn, and Power BI to visualize sensor behavior and potential failure patterns for maintenance strategy discussions.
- Explored and benchmarked machine learning approaches such as SVM, KNN and Logistic Regression using Scikit-learn on sample datasets to evaluate predictive maintenance applicability.

PROJECTS

Sales Data Transformation and Analysis | R, SQLite, Snowflake, ggplot2

Jan 2025 – Apr 2025

- Engineered an ETL pipeline using R, SQLite, and Snowflake schema to automate processing of 1M+ sales records, cutting manual preprocessing by 20% and accelerating data refresh cycles.
- Transformed OLTP data into OLAP structures and performed advanced wrangling to deliver actionable insights; built ggplot2 visualizations that improved forecast accuracy and guided non-technical stakeholders.

Predictive Modeling for Alzheimer's Onset | Python, Pytorch

Aug 2024 – Dec 2024

- Processed and transformed 5,000+ MRI brain scans with advanced pre-processing (scaling, noise reduction, augmentation) to overcome class imbalance and enable reliable analysis
- Designed and evaluated deep learning pipelines (CNNs & ViTs), delivering 85% AUC on balanced data and uncovering key insights on how pre-processing strategies drive diagnostic accuracy

Retail Insights Analysis | Excel, Power BI

Aug 2024 – Dec 2024

- Consolidated and standardized raw Excel sales files using Power Query to clean, format, and structure data; accelerated monthly revenue analysis and ensured consistent reporting across regional teams.
- Designed an interactive Power BI dashboard with layered KPIs, slicers, and dynamic filters; enabled executives to compare performance by region and derive timely, actionable insights.

NBA Playoffs Analysis | Python, Tableau, Data Cleaning

May 2024 – Aug 2024

- Automated daily scraping of 90+ playoff game stats with Python and optimized data cleaning using Pandas/NumPy, reducing manual effort by 90% and improving processing speed by 30% .
- Built a live interactive Tableau dashboard to visualize trends in scoring, player efficiency, and win probabilities, enabling dynamic data exploration throughout the playoffs..

TECHNICAL SKILLS

Programming Languages: Python, SQL (DDL, DML, joins, CTEs, window functions, query optimization), R, C, C++

Data Analysis: Pandas, NumPy, Data Cleaning, Data Wrangling, Statistical Analysis, Feature Engineering, Machine Learning (scikit-learn: Logistic Regression, KNN, SVM, Random Forest, Gradient Boosting, Neural Networks, Model Evaluation)

Data Visualization: Power BI, Tableau, Matplotlib, Seaborn

ETL & Data Processing: Alteryx, RESTful APIs, ETL Pipelines, Workflow & Reporting Automation

Databases & Cloud: MySQL, BigQuery, Snowflake, SQLite, AWS (S3, Lambda, Redshift), GitHub, Prompt Engineering

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- McKinsey Forward Program Graduate
- Tableau Data Analyst